

INTERNSHIP REPORT

Artificial Intelligence, Machine Learning and Deep Learning

Organization	Elevance Skills Technology
Track	AI, ML & Python
Name	Shalini Veronica
Course	B.Sc. (Hons.) Computer Science
College	RK Degree and PG College
University	Telangana University
Year	2026

1. Introduction

This report presents the work completed during the Elevance Skills Technology internship in Artificial Intelligence, Machine Learning and Python. The internship provided practical experience in developing, training, testing and integrating machine learning and computer vision applications.

2. Background

The internship focused on practical AI and computer vision applications using Python. The projects involved image classification, face recognition, attendance monitoring, animal detection, drowsiness detection, sign language recognition, vehicle colour detection and nationality-related classification.

3. Learning Objectives

The main objectives were to strengthen Python programming, understand machine learning and deep learning workflows, work with image datasets, train and test models, develop graphical user interfaces, use OpenCV, organize project files, and document completed applications.

4. Activities and Tasks

Seven projects were included in the internship submission:

1. Emotion Detection – Developed an application that detects facial emotions using a trained deep learning model and OpenCV.
2. Attendance System – Developed a face-based attendance application that identifies students, records attendance and integrates emotion information with time-based attendance handling.
3. Animal Detection – Developed an application for detecting multiple animals and identifying carnivorous animals using trained computer vision models.
4. Drowsiness Detection – Developed a system to classify a detected face as awake or sleeping using a trained image classification model.
5. Sign Language Detection – Developed a sign recognition application for signs including hello, thank you, yes, no and help, with image/video interaction.

6. Car Colour Detection – Developed a traffic-scene application to detect car colours, count cars and count people from images.

7. Nationality Detection – Included as an additional project as instructed by the internship instructor. The project contains Python training/testing scripts and a trained nationality-related classification model.

5. Skills and Competencies

Python programming, OpenCV, machine learning, deep learning, image processing, model training, model testing, dataset preparation, computer vision, GUI development, file and project organization, debugging, and technical documentation.

6. Feedback and Evidence

Evidence of the work includes the completed project source files, training and testing scripts, trained model files, graphical interfaces and the internship report. The projects were tested during development and organized into separate project folders for submission.

7. Challenges and Solutions

Challenges included organizing multiple project files, handling datasets and trained models, resolving Python and OpenCV environment issues, improving model training results, testing camera and image inputs, and separating final project files from earlier experimental files. These issues were addressed through debugging, repeated testing, dataset organization and refinement of the project structure.

8. Outcomes and Impact

The internship resulted in practical experience in building AI and computer vision applications. The completed projects provided experience with the workflow from dataset preparation and model training to testing, GUI development and documentation. The additional Nationality Detection project was also included as instructed.

9. Conclusion

The Elevance Skills Technology internship provided hands-on exposure to Artificial Intelligence, Machine Learning, Deep Learning and Python-based computer vision. Completing the seven projects improved practical programming, model development, testing, debugging and documentation skills and provided experience in developing application-oriented projects.

Declaration

I hereby declare that the projects and information presented in this internship report represent the work completed and organized for my Elevance Skills Technology internship submission.